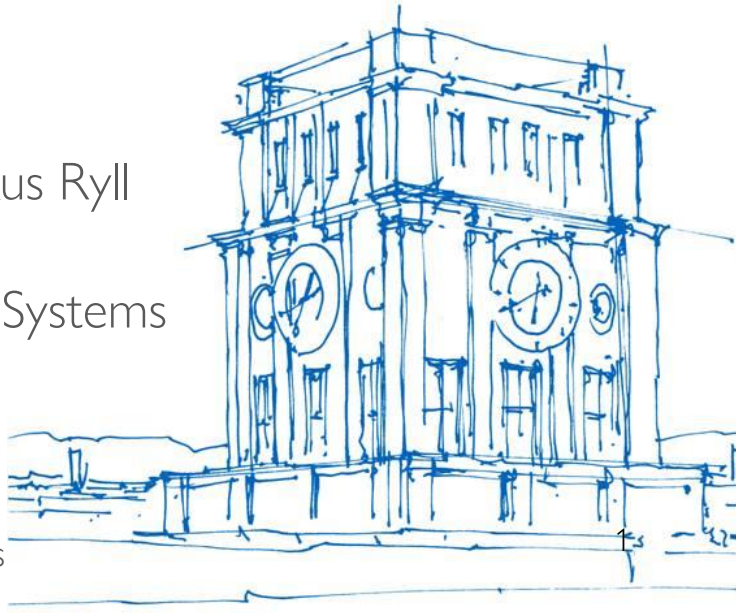
Engineering Mechanics I

Prof. Dr.-Ing. Markus Ryll

Autonomous Aerial Systems



Lecturing Team



Yuxia
Yuan



Markus
Ryll



Sidar
Yilmaz

Markus Ryll - Biography



Since Sept. 2020: **Professor for Autonomous Aerial Systems**,
at Technical University Munich, München, Germany

- Highspeed manipulation in unknown environments
- Human-aerial robot collaboration

Jan. 2018 – Jul. 2020: **Postdoctoral Researcher** in the Robust Robotics Group,
at Massachusetts Institute of Technology, Cambridge, USA

- Real-time Trajectory planning in unknown environments
- Semantics based trajectory planning

Sept. 2014 – Mar. 2017: **Postdoctoral Researcher** in the RIS team,
at CNRS LAAS, Toulouse, France

- Towards Aerial physical interaction

Sept. 2010 – Jul. 2014: **Ph.D. Control Theory** in the Autonomous Robotics Group
at MPI for Biological Cybernetics, Tübingen, Germany

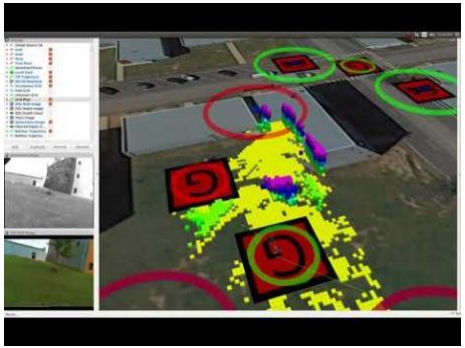
- Design and control of a fully actuated aerial vehicle

Research - Autonomous Aerial Systems



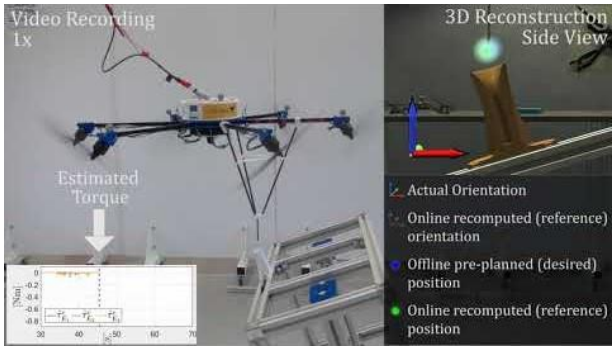
Autonomous Flying

- High Autonomous Speed Flight
- Search and Rescue



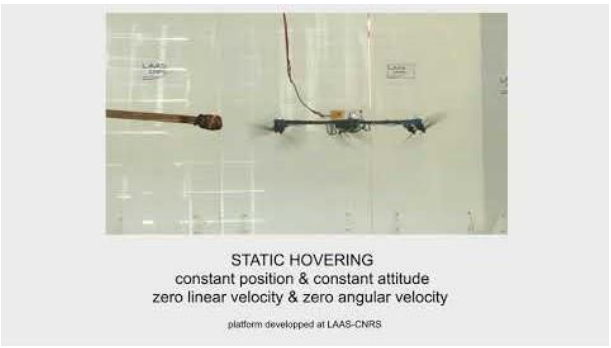
Aerial Manipulation

- Grasping / Transport.
- Human-Robot Interaction



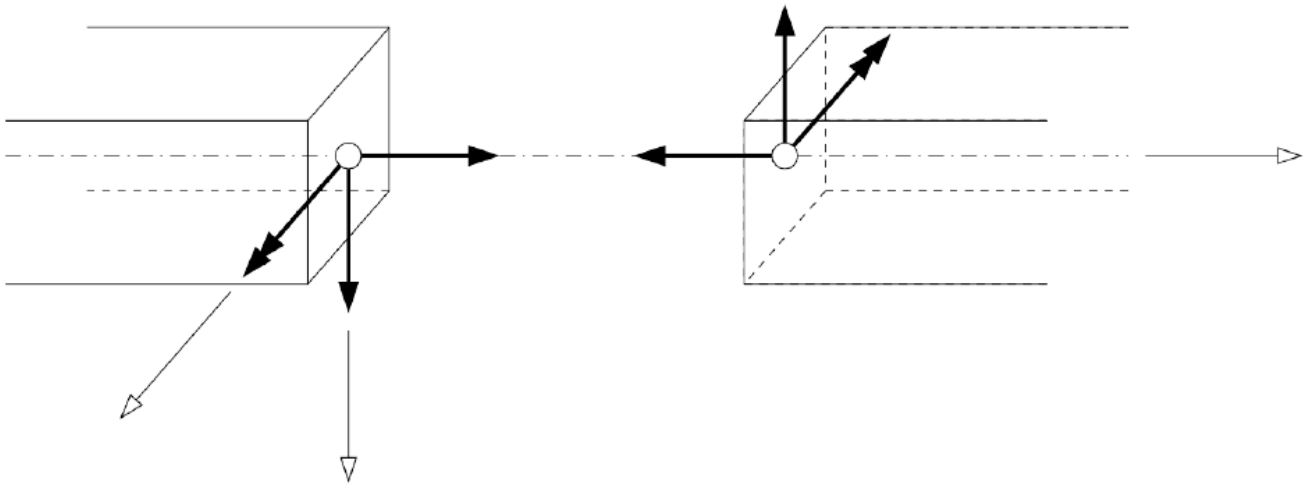
Transportation / Safety

- Robust Control
- Failure Redundancy



Engineering Mechanics I

General information



Lectures in Mechanics

- Semester 1 – 4
  - Engineering Mechanics I (**Fundamentals**), II & III
  - Fluid Mechanics I – II
  - Aerospace Structures and Elements I – II
- Semester 5 – 6
  - Modelling Elective: Computational Solid Mechanics in Aerospace
  - General Elective

Exams in Mechanics

- Exams
  - EM I (WS) Exam: written, 90min
  - EM II (SS) Exam: written, 90min

- EM I (Fundamentals): **Needs to be completed by the end of the 2nd semester**

Lecture Dates

Tuesday	14.10.2025	09:45
Thursday	16.10.2025	08:45
Tuesday	21.10.2025	09:45
Thursday	23.10.2025	08:45
Tuesday	28.10.2025	09:45
Thursday	30.10.2025	08:45
Tuesday	04.11.2025	09:45
Thursday	06.11.2025	08:45
Thursday	13.11.2025	08:45
Tuesday	18.11.2025	09:45
Thursday	20.11.2025	08:45
Tuesday	25.11.2025	09:45
Thursday	27.11.2025	08:45
Tuesday	02.12.2025	09:45
Tuesday	09.12.2025	09:45
Thursday	11.12.2025	08:45
Tuesday	16.12.2025	09:45
Thursday	18.12.2025	08:45
Tuesday (Christmas Lecture)	23.12.2025	09:45
Winter Break		
Thursday	08.01.2026	08:45
Tuesday	13.01.2026	09:45
Thursday	15.01.2026	08:45
Tuesday	20.01.2026	09:45
Thursday	22.01.2026	08:45
Tuesday	27.01.2026	09:45
Thursday	29.01.2026	08:45
Tuesday	03.02.2026	09:45
Thursday	05.02.2026	08:45

Lectures and Exercises for EM I



Confucius, Chinese Philosopher
551 – 479 BC

"Tell me and I will forget,

- Lecture

show me and I may remember;

- Central Exercise

involve me and I will understand."

- Exercise, Assignments, Matlab-Exercises

Lectures and Exercises for EM I

Lecture

- Tuesday 9.45 am – 11.15 am
- Thursday 08.45 am – 09.30 am

- The lecture is authoritative for the scope of the material.

- Script and completed slides on learning platform www.moodle.tum.de

Lectures and Exercises for EM I



Central Exercise

- Every week a new central exercise (in person, Tuesday)

- Pre-calculating Exercises
- Exemplary, task oriented
- Matlab required: https://wiki.in.tum.de/Informatik/Helpdesk/MatlabInstallieren

Lectures and Exercises for EM I

Central Exercise

- Weekly assignment sheets on learning platform www.moodle.tum.de (Start: 2nd week)
- Possibilities on **Piazza** and in **tutor consultation hours** to ask questions about the in-depth exercises
- Supervision by tutors (students of higher semesters)
- Answering questions about the **completed (!!)** exercise sheets

➢ **Goal: Feedback on own knowledge level in mechanics**

Lectures and Exercises for EM I

Tutor consultations

- Voluntary additional offer
- No registration required

- Dates: (weekly, starting 2nd week): Wednesdays @ 15.00am – 17.00

Lectures and Exercises for EM I

Homework

- Assignments about every 2 weeks in addition to the exercise sheet (start 4th week)
- Processing time: 2 weeks

➢ Goal: Feedback and continuous learning instead of "cramming" before the exam.

Lectures and Exercises for EM I

Midterm Exam (Voluntary additional offer)

- Dress rehearsal for the EM I exam
- Expected in the last week of the semester
- Exact date and further information will be announced in the course

➢ Goal: Dry run for an "exam atmosphere"

Updates and Downloads

www.moodle.tum.de

- Download center (script, videos, lecture slides, exercise sheets, solutions, ...)
- Notifications are made in the **Messages** section
- With default settings you will be informed about new messages by mail

1. Automatic registration, if you have registered for the lecture EM 1 in TUMonline.
2. Login with MyTUM email address and password
3. Access to all documents under *My Area* → *Current Courses* → *Engineering Mechanics I*

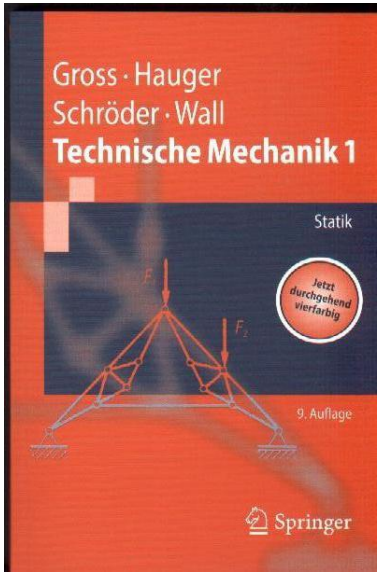
Contact

- Autonomous Aerial Systems: Department of Aerospace and Geodesy (2nd floor)
  Lise-Meitner-Str. 9, 85521 Ottobrunn



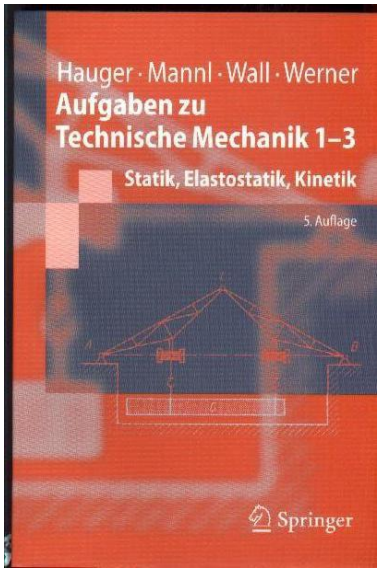
- Office Hours: By agreement

1.2. Complementary Literature



- D. GROSS, W. HAUGER, J. SCHRÖDER, W. A. WALL und N. RAJAPAKSE: Engineering Mechanics Band 1-3, Springer Verlag, Berlin, 2009 - 2011.
- J.H. WILLIAMS: Fundamentals of Applied Dynamics, John Wiley & Sons, 1996.
- K. MAGNUS und H.H. MÜLLER-SLANY: Grundlagen der Technischen Mechanik, Teubner Verlag, Stuttgart 2005.

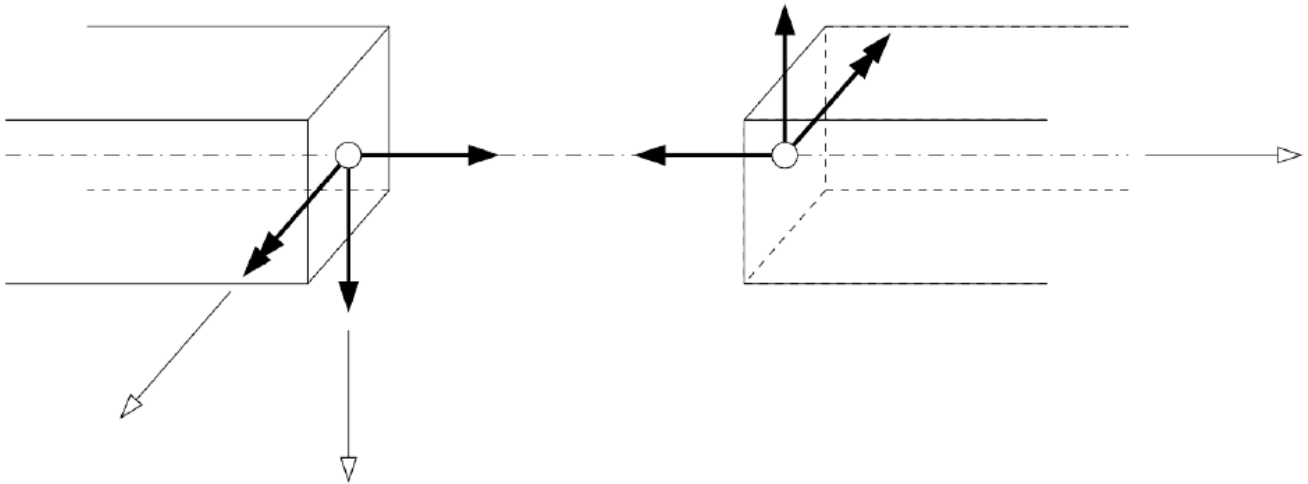
1.3. Exercises



- W. HAUGER, V. MANNL, W. A. WALL und E. WERNER: Aufgaben zur Technischen Mechanik 1: Statik, Elastostatik, Kinetik, Springer Verlag, Berlin, 2008.

Engineering Mechanics I

Lecture 1



2.1. The task of mechanics

- Mechanics is a branch of physics
- Mechanical state under the influence of forces
- Movement of the body corresponds to the temporal course of the change of location of the body points
- **Rest as special case included (statics)**
- Frame of (Euclidean) space and time
- Body is material continuum, carrier of physical quantities

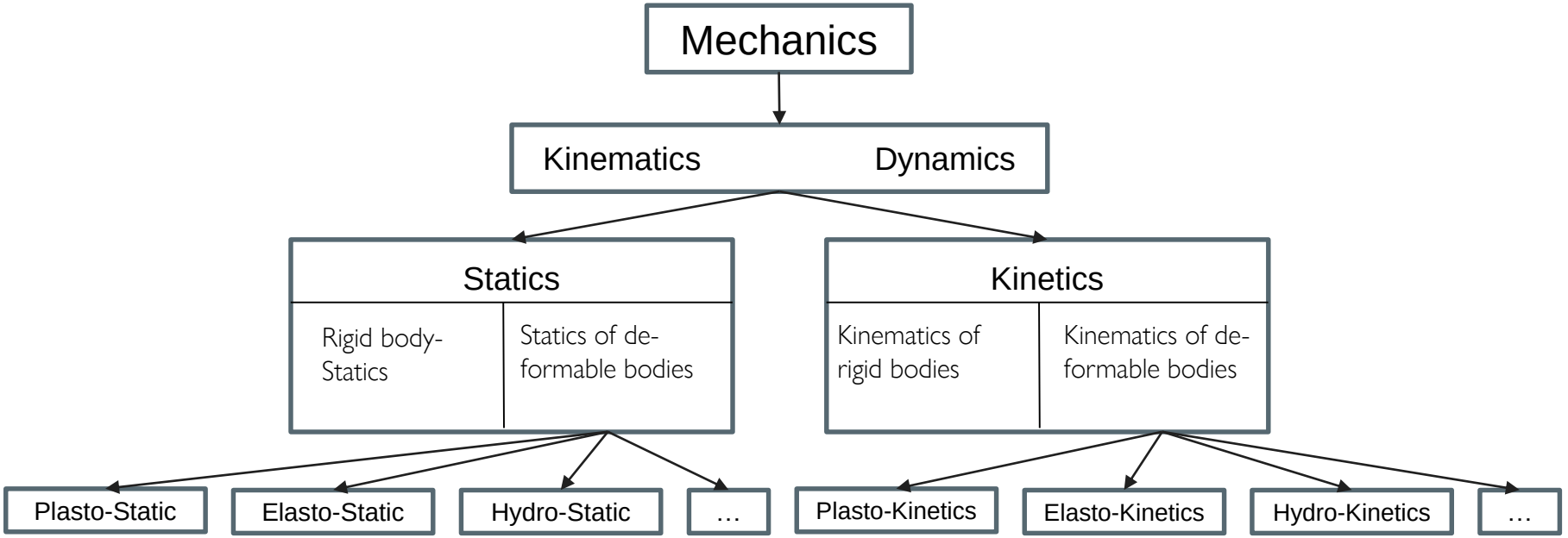
2.1. The task of mechanics

- Scientific laws, based on observations
- Model expressed in the language of mathematics
- Never complete
- Model of reality

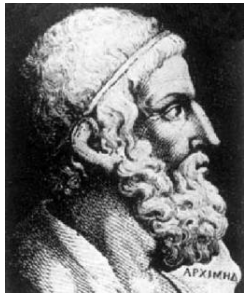
Engineering Mechanics

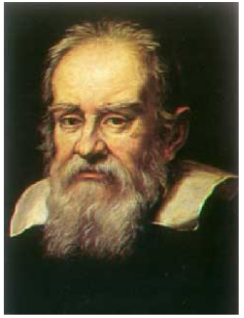
- Draws conclusions from physical mechanics for technical systems
- Predicts the behavior of technical systems based on theoretical considerations

2.2 Classification of Mechanics



2.3 Historical development





- ARCHIMEDES (287-212 v. Chr.):
  Lever law, center of gravity, hydrostatics

- LEONARDO DA VINCI (1452-1519)
  Mechanical Systems (Robots, planes,…)



- GALILEI (1564–1642):
  Free fall, Projectile motion

- NEWTON (1643-1727):
  Kinetics of mass points, link between mass, velocity, momentum and force
  "Philosophia naturalis principia mathematica" (Print 1687)



2.3 Historical development

- Family BERNOULLI (17. and 18. century):
  Beam, pipe flow
- EULER (1707–1783):
  Rigid Body Kinetics
- D'ALEMBERT (1717–1783):
  Inertia
- LAGRANGE (1736–1813):
  Calculus of variation

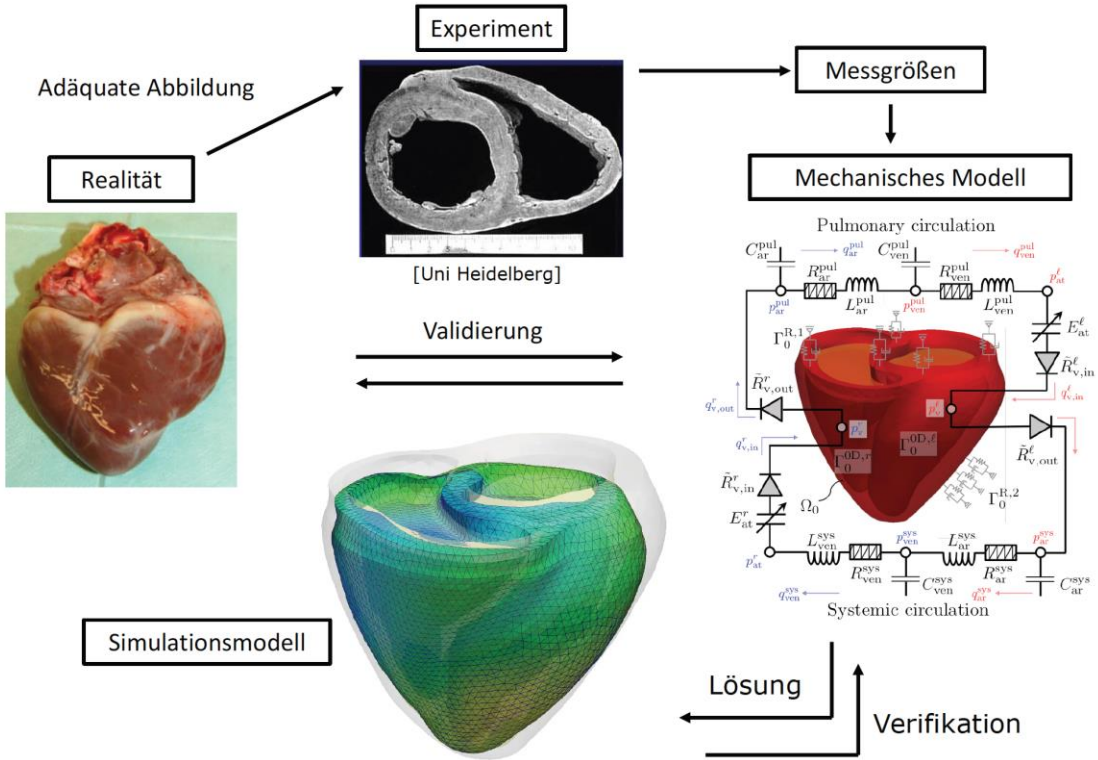








3. Modeling in Mechanics



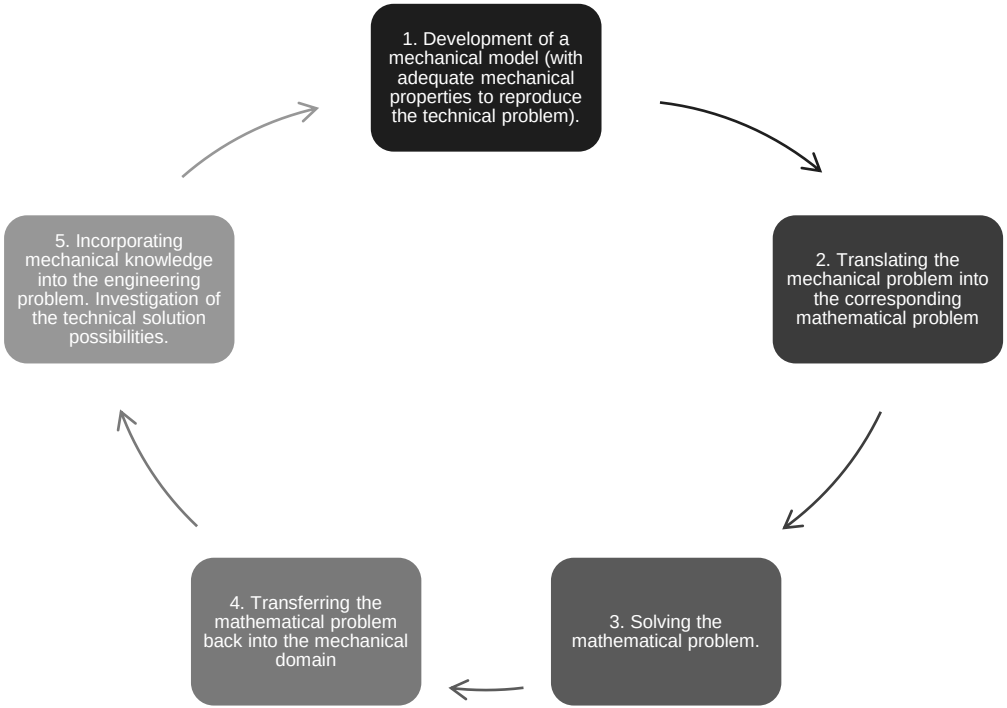
3. Modeling in Mechanics







3. Modeling in Mechanics



3. Modeling in Mechanics

Increasingly important role in everyday engineering

Solution of mathematical model problems by computer oriented methods increasingly easier.

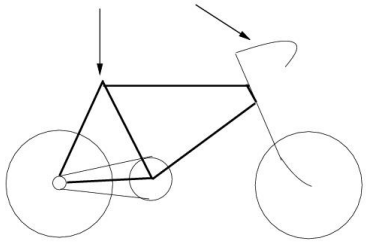


- Exact **description** of the problem
- Selection of the **solution method**
- Correct interpretation of the results as a solution to the model problem
- Quantification of uncertainties and assessment of model quality

3. Modeling in Mechanics

- Models should, with reasonable modeling and computational effort, capture the the essential properties of the real or virtual technical system with reasonable modeling and computational effort

- Good models are **as simple as possible and as complex as necessary**.





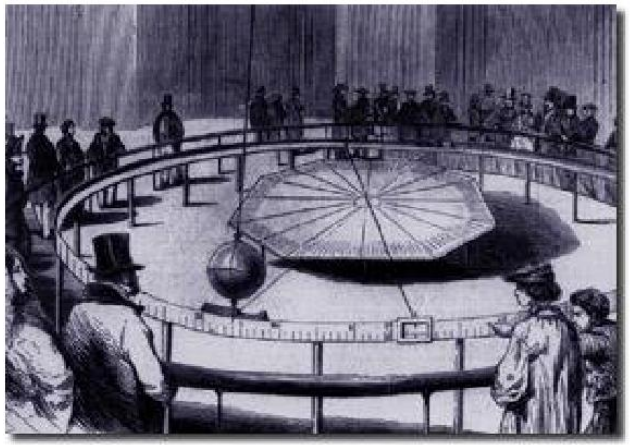
3. Modeling in Mechanics

- System properties are attributed to scalar and vector model parameters
- Engineering units are dimensionally related
  - e.g. Dimension = distance / time
  - Unit: m / s
- Different orders of magnitude lead to different units of measurement - include in calculation!
- Coherent system of units: International System (SI)

Quiz

Which physical effects should be considered when building a model for a pendulum? Rank them in order of importance.

a) Air friction
b) Gravitational force
c) Friction in the suspension system
d) Weight of the rope / rod

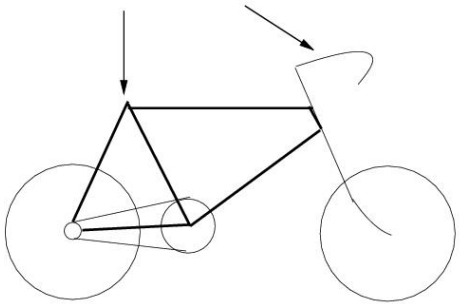


Foucault's pendulum

Quiz

Match the model of a bicycle shown to the two tasks.

a) Calculation of wheel contact forces
b) Strength verification for bicycle frame





Mit freundlicher Genehmigung www.dyne.ch